When Does PEFT Help under Synthetic Document Noise? A Controlled Study of Gating, Capacity, and Robustness

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Abstract

Parameter-efficient fine-tuning (PEFT) is widely used to adapt pretrained multimodal transformers to downstream document-understanding tasks. However, it remains unclear whether PEFT mechanisms designed to respond to input corruption improve task performance because they identify corrupted evidence or simply increase model capacity. We present a controlled diagnostic study of gated PEFT for token-level entity recognition on the FUNSD dataset using LayoutLMv3 and Low-Rank Adaptation (LoRA). All methods are evaluated under a matched training protocol across five random seeds, identical optimizer configurations, and controlled synthetic text and layout corruptions. A hand-crafted quality-weighting and gradient-masking method, SALLoRA, consistently fails to improve over standard LoRA and degrades macro-F1 by up to 3.09 percentage points under severe OCR noise. A simpler method, GateConcat, concatenates a learned scalar token feature with the LoRA hidden state before classification, improving noisy macro-F1 over LoRA by +1.51 percentage points under 70% character substitution ($p = 0.003$) and +2.30 points under 70% OCR confusion ($p = 0.002$). However, a constant-gate capacity control ($g \equiv 0.5$) captures a substantial portion of the improvement (+0.74 points on OCR), and the difference between GateConcat and the constant control is not statistically significant ($p = 0.21$). Furthermore, the learned gate does not identify corrupted tokens: its corruption-detection AUC is approximately 0.52 and its token-level correlation with corruption is -0.025. The gain also fails to transfer to layout box-swap noise (-0.20 points). Our findings demonstrate that modest robustness improvements under severe synthetic text corruption can arise from classifier-head capacity or label-salience routing rather than learned corruption detection, underscoring the necessity of capacity-matched controls when evaluating noise-aware PEFT architectures.

1 Introduction

Document AI systems operating in enterprise environments must process visual and textual inputs that are frequently degraded by scanner artifacts, optical character recognition (OCR) substitutions, missing text fragments, and spatial alignment errors [1, 2]. In token-level tasks such as form entity recognition, structured input noise alters the local input features while leaving the underlying semantic entity labels intact. Developing adaptation methods that maintain performance under such noise is critical for robust document processing.

Recent advances in document understanding rely heavily on multimodal transformer architectures that jointly model textual, visual, and spatial modalities [2–4]. LayoutLMv3, for instance, unifies text and image masking objectives during pretraining and achieves strong benchmark performance [2]. Adapting these multi-hundred-million parameter backbones via full fine-tuning, however, incurs high computational and storage costs. Parameter-efficient fine-tuning (PEFT) techniques, particularly Low-Rank Adaptation (LoRA) [5] and adapter modules [6], mitigate this burden by freezing the pretrained backbone weights and updating only low-rank parameter decomposition matrices inserted into self-attention projections.

Despite the efficiency of standard LoRA, it offers no explicit mechanism to handle degraded or corrupted input tokens. This has motivated noise-aware and uncertainty-gated adaptation strategies designed to detect unreliable features, route token representations, reweight training losses, or down-scale updates from corrupted inputs [7, 8]. A core assumption underlying such architectures is that a learned quality or corruption-routing signal enables the model to bypass or suppress noisy evidence. However, this assumption is not necessarily valid for document entity recognition: a corrupted token may still retain partial contextual evidence, whereas an explicit noise predictor may learn features irrelevant to downstream entity classification.

In this work, we present a controlled empirical study examining whether this assumption holds for document entity recognition. Using LayoutLMv3-base with LoRA on the FUNSD dataset [1], we conduct matched evaluations comparing standard LoRA against several noise-aware variants: a hand-crafted quality-weighting and gradient-masking recipe (SALLoRA), residual routing mechanisms, auxiliary corruption prediction, and a lightweight feature-concatenation approach called GateConcat.

GateConcat derives a scalar feature from the base representation, adapted LoRA representation, feature disagreement norm, and spatial bounding box, concatenating this scalar with the LoRA hidden state prior to linear classification. While GateConcat produces modest, statistically significant macro-F1 improvements under severe synthetic text corruption, introducing capacity-matched controls reveals a more nuanced reality.

A constant-gate control ($g \equiv 0.5$) with identical classifier input dimensionality also outperforms standard LoRA, and the performance gap between GateConcat and this constant control is not statistically significant.

Our study investigates the empirical mechanism behind these gains by addressing five concrete questions:

1. Do hand-crafted or supervised quality-weighting mechanisms improve document NER under controlled synthetic corruption?
2. Does GateConcat consistently improve noisy macro-F1 over an architecture-matched LoRA baseline?
3. Does the learned token gate significantly outperform a capacity-matched constant scalar control?
4. Does the learned gate explicitly detect or track input token corruption?
5. Do robustness gains under text corruption transfer to spatial perturbations?

Our primary findings are summarized as follows:

- Hand-crafted quality weighting and gradient masking (SALLoRA) fail to outperform standard LoRA, degrading noisy macro-F1 by up to -0.0309 under OCR confusion.
- GateConcat achieves modest, statistically significant noisy macro-F1 gains under severe text noise (+1.51 pp under 70% substitution, $p = 0.003$; +2.30 pp under 70% OCR confusion, $p = 0.002$).
- A constant-gate capacity control ($g \equiv 0.5$) captures part of this benefit (+0.74 pp on OCR noise), and GateConcat does not significantly outperform this control ($p = 0.21$).
- The learned gate does not act as a corruption detector (AUC ≈ 0.52; correlation with corruption $r = -0.025$). Auxiliary supervision forces high corruption detection (AUC ≈ 0.91) but fails to improve downstream task F1.
- Gains are text-specific and do not transfer to layout perturbations, yielding a slight degradation under 50% bounding-box swaps (-0.002).

Rather than proposing a novel noise-aware algorithm, this paper provides a diagnostic case study demonstrating that PEFT robustness gains under synthetic text noise can easily be misattributed to learned corruption routing when they are in fact driven by classifier-head capacity and token-salience interaction.

2 Related Work

2.1 Multimodal Document Understanding

Visually rich document understanding (VRDU) requires integrating textual semantics, visual document images, and two-dimensional spatial layout structures [3]. Early document processing pipelines processed these signals independently through isolated OCR, vision, and NLP modules. Transformer-based architectures unified these streams by introducing spatial bounding-box position embeddings alongside textual token embeddings [3]. LayoutLMv3 further unified pretraining by applying masked language modeling on text tokens and masked image modeling on visual patch tokens [2]. While these backbones achieve state-of-the-art results on form understanding benchmarks like FUNSD [1], their behavior under structured input noise remains insufficiently characterized.

2.2 Parameter-Efficient Fine-Tuning (PEFT)

Parameter-efficient fine-tuning adapts large pretrained models while modifying only a small fraction of the model parameters. Adapter-based methods insert trainable bottleneck layers between transformer sub-layers [6]. Low-Rank Adaptation (LoRA) parameterizes weight updates through low-rank decomposition matrices $W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA$, typically applied to multi-head self-attention query and value projection matrices [5]. Subsequent variants optimize rank allocation or learning rates. However, standard PEFT methods assume clean downstream training and evaluation inputs, leaving open the question of how adapter architectures perform under structured input degradation.

2.3 Robust PEFT and Gating Mechanisms

Handling noisy inputs in neural networks traditionally involves sample reweighting, robust loss functions such as focal loss or label smoothing [7, 8], or dynamic feature gating. In PEFT, gating modules are frequently proposed to dynamically route representations or scale adapter outputs based on input quality or task context. Recent work on noise-aware and robust PEFT has explored federated LoRA under label noise, sensitivity-aware tuning, and uncertainty-guided adaptation [9, 10]. These methods primarily target federated settings and focus on label noise or client heterogeneity rather than input token corruption in document understanding. Our study complements this line of work by examining input-level corruption in a centralized document NER setting and testing whether gating improves performance beyond capacity-matched controls.

2.4 Capacity Controls in PEFT

A recurring challenge in PEFT research is disentangling architectural effects from capacity effects. Adding adapter modules, prompt tokens, or gating branches changes the number of trainable parameters and the effective capacity of the classifier head. Prior work on visual prompting and embedded prompt tuning highlights that even small prompt modules can substantially change adaptation behavior [11, 12]. However, few studies systematically compare learned gating mechanisms against constant or randomized controls with matched dimensionality. Our work addresses this gap by introducing constant-gate and simplified-scalar controls matching the classifier-input dimensionality of GateConcat.

3 Experimental Protocol

3.1 Dataset and Task Formulation

We conduct all experiments on the FUNSD dataset [1], a standard benchmark for form understanding consisting of 199 fully annotated scanned forms (149 training, 50 testing). The task is framed as sequence labeling over OCR tokens into semantic entity categories (Header, Question, Answer, Other). Evaluation metrics include entity-level macro-F1 and token-weighted F1. We prioritize macro-F1 to ensure equal sensitivity across frequent and rare entity classes.

To evaluate robustness, we introduce controlled synthetic corruptions to test documents while preserving ground-truth semantic labels:

- **Character Substitution**: Replaces characters with random alphanumeric characters at probability $p \in \{0.0, 0.3, 0.5, 0.7\}$.
- **OCR Confusion**: Simulates realistic OCR errors by replacing characters with visually or acoustically confusion-prone counterparts (e.g., 'o' $\rightarrow$ '0', 'l' $\rightarrow$ '1') at $p = 0.7$.
- **Token Insertion / Deletion**: Randomly inserts distractor tokens or deletes existing tokens at $p = 0.7$.
- **Bounding-Box Swap**: Randomly swaps spatial bounding box coordinates between token pairs at probability $p = 0.5$.

3.2 Backbone and Adapter Configuration

All methods utilize `microsoft/layoutlmv3-base` (125M parameters) as the frozen underlying transformer backbone. Trainable LoRA modules ($r = 16, \alpha = 32$, dropout $= 0.1$) are inserted into the self-attention 'query' and 'value' projection matrices. A shared classifier head projects token representations to target label logits.

3.3 Matched Training Hyperparameters

To eliminate hyperparameter tuning bias, all methods are trained using an identical configuration [13]:

- **Optimizer**: AdamW [13] with initial learning rate $\eta = 2 \times 10^{-4}$, weight decay $\lambda = 0.01$.
- **Schedule**: 15 epochs, batch size 4, linear learning rate warmup over the first 10% of total steps.
- **Objective**: Standard unweighted cross-entropy loss (unless loss variant is explicitly specified).
- **Random Seeds**: Five independent runs per setting using seeds $\mathcal{S} = \{42, 123, 2024, 456, 789\}$.

3.4 Statistical Methodology

For paired method comparisons across identical random seeds, we evaluate per-seed macro-F1 differences $\Delta_s = \text{MacroF1}_s(M_A) - \text{MacroF1}_s(M_B)$. We report two-tailed paired Student's t-test p-values alongside non-parametric Wilcoxon signed-rank test p-values. With $S = 5$ seeds, the minimum attainable two-sided Wilcoxon p-value is 0.0625; thus, paired t-tests serve as the primary metric for continuous significance reporting.

4 Methods

4.1 Standard LoRA Baseline

Let $h_i^{\text{base}} \in \mathbb{R}^H$ denote the token output representation from the frozen LayoutLMv3 backbone ($H = 768$), and $h_i^{\text{LoRA}} \in \mathbb{R}^H$ denote the adapted hidden state after applying self-attention LoRA updates [5]. The baseline classifier computes logits $z_i \in \mathbb{R}^C$:

$$z_i = W_{\text{cls}} h_i^{\text{LoRA}} + b_{\text{cls}}, \quad W_{\text{cls}} \in \mathbb{R}^{C \times H} \tag{1}$$

4.2 SALLoRA (Static Quality-Weighted Baseline)

SALLoRA evaluates a hand-crafted robustness recipe combining heuristic token quality scoring, gradient masking, and layer-wise scaling. A token quality score $q_i \in [0, 1]$ is computed from text character entropy and bounding-box aspect ratios. The task loss is weighted by q_i, and gradients for adapter parameters corresponding to tokens with $q_i < 0.3$ are zeroed during backpropagation.

4.3 Residual Routing

Residual routing models compute a dynamic token gate $g_i \in (0, 1)$ to scale the LoRA residual update:

$$h_i^{\text{route}} = h_i^{\text{base}} + g_i \cdot \left(h_i^{\text{LoRA}} - h_i^{\text{base}}\right) \tag{2}$$

Logits are then computed as $z_i = W_{\text{cls}} h_i^{\text{route}} + b_{\text{cls}}$. Variants are evaluated with pure task loss and with an auxiliary binary cross-entropy loss $\mathcal{L}_{\text{aux}}$ driving $g_i \rightarrow 0$ on synthetically corrupted tokens.

4.4 GateConcat

GateConcat constructs a rich scalar feature $g_i \in (0, 1)$ incorporating information from both the frozen backbone and adapted representations:

$$\Delta_i = h_i^{\text{LoRA}} - h_i^{\text{base}} \tag{3}$$

$$v_i = \left[h_i^{\text{base}} \,;\, \Delta_i \,;\, \|\Delta_i\|_2 \,;\, \text{bbox}_i\right] \in \mathbb{R}^{2H+5} \tag{4}$$

$$g_i = \sigma\left(W_2 \cdot \text{GeLU}\left(W_1 v_i + b_1\right) + b_2\right) \tag{5}$$

where $\text{bbox}_i = (x_0, y_0, x_1, y_1, \text{area})$ contains normalized spatial coordinates. Rather than suppressing adapter outputs, GateConcat appends g_i directly to the classifier input:

$$z_i = W_{\text{concat}} \left[h_i^{\text{LoRA}} \, ; \, g_i \right] + b_{\text{concat}}, \quad W_{\text{concat}} \in \mathbb{R}^{C \times (H+1)} \tag{6}$$

Training uses standard unweighted cross-entropy on entity labels without auxiliary corruption supervision.

4.5 Capacity Controls

To isolate the effect of learned gating from classifier-head dimensionality expansion ($H \rightarrow H + 1$), we introduce two control baselines:

- **GateConcatFrozen**: Replaces g_i with a constant scalar value $g_i \equiv 0.5$. The classifier receives an identical $H + 1$ input dimension, but the appended feature carries zero token-specific information.
- **ScalarConcat**: Generates a scalar $g_i = \sigma(W_{\text{proj}} h_i^{\text{LoRA}})$ from a minimal single-layer projection without feature differences or bounding-box inputs.
- **GateConcatAux**: Trains the GateConcat architecture with an added soft auxiliary loss ($\lambda = 0.1$, warmup over 3 epochs) forcing g_i to predict synthetic token corruption.

5 SALLoRA: A Failed Baseline

We first evaluate the static SALLoRA baseline across multiple loss function configurations under controlled synthetic corruption on FUNSD. As shown in Table 1, SALLoRA consistently underperforms standard LoRA.

Under 70% character substitution with cross-entropy, SALLoRA degrades macro-F1 by -0.0217. Under 70% OCR confusion, the performance penalty reaches -0.0309. While focal loss yields a minor positive delta (+0.0076), this is driven by loss function reweighting rather than the SALLoRA gating mechanism.

These results demonstrate that hand-crafted quality down-weighting is counterproductive for document NER. Corrupted OCR tokens frequently retain critical contextual and spatial cues necessary for boundary detection; explicitly suppressing these tokens deprives the classifier of essential evidence.

Table 1: Controlled comparison between Standard LoRA and SALLoRA on FUNSD across 5 seeds. Values represent Macro-F1 and Weighted-F1 on noisy test sets.

Corruption / Loss	LoRA W	SAL W	LoRA M	SAL M	Δ Macro
Sub @ 70% + CE	0.7799	0.7772	0.7233	0.7016	-0.0217
Sub @ 70% + LS	0.7826	0.7814	0.7283	0.7035	-0.0248
Sub @ 70% + Focal	0.7719	0.7782	0.7180	0.7256	+0.0076
OCR @ 70% + CE	0.8009	0.7904	0.7414	0.7105	-0.0309
Box-Swap @ 50% + CE	0.8135	0.8138	0.7505	0.7473	-0.0031

6 GateConcat: Empirical Performance

Evaluating GateConcat across varying character substitution rates reveals a clear severity-dependent trend (Table 2).

At low noise levels (0% and 30%), GateConcat performs identically to standard LoRA. Performance gains emerge at 50% corruption (+0.0064) and become statistically significant at 70% character substitution (+0.0151 macro-F1 gain, paired t-test $p = 0.003$). Under 70% substitution, all five random seeds yield positive deltas for GateConcat over LoRA.

Table 2: Substitution rate sweep on FUNSD (Mean ± Std across 5 seeds). Comparisons show clean and noisy test macro-F1.

Rate	LoRA Noisy	GateConcat Noisy	Δ Noisy	p-value
0%	0.7511 ± 0.0068	0.7512 ± 0.0084	+0.0001	0.982
30%	0.7457 ± 0.0051	0.7447 ± 0.0176	-0.0010	0.891
50%	0.7300 ± 0.0187	0.7364 ± 0.0147	+0.0064	0.312
70%	0.7312 ± 0.0104	0.7463 ± 0.0130	**+0.0151**	**0.003**

7 Capacity Controls and Interpretation

To determine whether GateConcat's success stems from dynamic feature gating or simple capacity expansion, we compare GateConcat against GateConcatFrozen, ScalarConcat, and GateConcatAux under identical 70% substitution (Table 3) and 70% OCR confusion protocols (Table 4).

Table 3: Capacity and auxiliary loss controls under 70% character substitution on FUNSD (5 seeds).

Method Architecture	Clean M	Noisy M	Noisy W	Gate AUC
Standard LoRA	0.7203 ± 0.0089	0.7312 ± 0.0104	0.7887	—
GateConcat	$\mathbf{0.7226 \pm 0.0149}$	$\mathbf{0.7463 \pm 0.0130}$	**0.8016**	0.521
GateConcatFrozen ($g \equiv 0.5$)	0.7221 ± 0.0099	0.7336 ± 0.0203	0.7872	—
ScalarConcat	0.7030 ± 0.0224	0.7246 ± 0.0220	0.7919	0.509
GateConcatAux ($\lambda = 0.1$)	0.7106 ± 0.0124	0.7318 ± 0.0072	0.7869	**0.909**

Table 4: Capacity control evaluation under 70% OCR confusion on FUNSD (5 seeds).

Method	Clean Macro	Noisy Macro	Δ vs LoRA	p-value
Standard LoRA	0.7333 ± 0.0112	0.7314 ± 0.0105	—	—
Frozen ($g \equiv 0.5$)	0.7431 ± 0.0120	0.7388 ± 0.0215	+0.0074	0.384
GateConcat	$\mathbf{0.7472 \pm 0.0115}$	$\mathbf{0.7544 \pm 0.0108}$	**+0.0230**	**0.002**

As detailed in Table 3 and Table 4: 1. **Frozen Baseline Beats LoRA**: Fixing $g \equiv 0.5$ improves noisy macro-F1 over standard LoRA by +0.0024 on substitution @70% and +0.0074 on OCR @70%. 2. **GateConcat vs. Frozen is Non-Significant**: While GateConcat achieves higher mean performance than Frozen (+0.0127 on substitution, +0.0156 on OCR), paired statistical testing reveals that these differences are not statistically significant ($p = 0.336$ for substitution @70%; $p = 0.210$ for OCR @70%). 3. **ScalarConcat Fails**: Appending a naive scalar projection degrades performance

(0.7246 macro-F1), showing that arbitrary scalar additions do not automatically improve classification.

These capacity controls demonstrate that expanding classifier input dimensionality ($H \rightarrow H + 1$) accounts for a notable fraction of the observed robustness gain. The hypothesis that learned token gating significantly outperforms a capacity-matched constant control cannot be confirmed at $n = 5$ seeds.

8 Corruption Detection vs. Task Robustness

A common assumption in noise-aware architectures is that improving a model's ability to detect corrupted inputs translates directly to downstream task robustness. To test this, GateConcatAux was trained with an explicit auxiliary binary cross-entropy objective ($\lambda = 0.1$) penalizing gate prediction errors relative to true token corruption status.

As shown in Table 3, auxiliary training successfully forces the gate to detect corruption, boosting corruption AUC from 0.521 to **0.909**. However, this high detection capability yields zero downstream NER benefit: GateConcatAux obtains 0.7318 noisy macro-F1, performing identically to standard LoRA (0.7312) and significantly worse than unassisted GateConcat (0.7463).

This result establishes a critical decoupling: **detecting synthetic input corruption is fundamentally distinct from learning robust task representations**. Forcing adapter modules to track noise patterns distracts model capacity from semantic entity recognition.

9 Transfer Tests across Noise Modalities

To test the operational boundary of GateConcat, we evaluate transfer performance across diverse text corruption types and spatial layout perturbations (Table 5).

Table 5: Transfer performance of GateConcat vs. Standard LoRA across synthetic text and layout corruptions on FUNSD (5 seeds).

Corruption Modality	LoRA Noisy	Concat Noisy	Δ Macro	p-value
OCR Confusion @ 70%	0.7314	0.7544	**+0.0230**	**0.002**
Deletion @ 70%	0.7302	0.7501	+0.0199	0.073
Insertion @ 70%	0.7371	0.7445	+0.0074	0.134
Box-Swap @ 50%	0.7505	0.7485	-0.0020	0.615

GateConcat demonstrates positive transfer across all synthetic text noise types, achieving gains on OCR confusion (+0.0230, $p = 0.002$) and deletion (+0.0199, $p = 0.073$). However, under spatial layout corruption (50% bounding-box swap), GateConcat yields a slight negative delta (-0.0020). The mechanism is strictly text-specific and fails to mitigate structural layout noise.

10 Gate Behavior Analysis

To analyze what the GateConcat scalar g_i learns in the absence of auxiliary supervision, we extract token-level gate diagnostics on the noisy test set for OCR @ 70% (Seed 42).

Table 6: Token-level gate correlation statistics for GateConcat under 70% OCR confusion (Seed 42, $N = 18,432$ test tokens).

Evaluated Token Property	Correlation with Gate (g)
Token Corrupted Flag (0 / 1)	**-0.025**
LoRA Disagreement Norm ($\|\Delta\|_2$)	-0.191
Bounding Box Area	-0.274
Token Sequence Position	-0.262
Prediction Correctness (0 / 1)	-0.041

The empirical correlation between g_i and true token corruption is -0.025, confirming that unassisted GateConcat does not learn to detect noise. Mean gate values for clean tokens (0.371 ± 0.11) and corrupted tokens (0.354 ± 0.12) are nearly identical.

Instead, g_i exhibits strong systematic variation across ground-truth semantic entity labels:

- **Header Tokens**: Mean $g = 0.824 \pm 0.08$
- **Question Tokens**: Mean $g = 0.412 \pm 0.11$
- **Answer Tokens**: Mean $g = 0.355 \pm 0.10$
- **Other / Background Tokens**: Mean $g = 0.072 \pm 0.04$

The learned gate functions as an implicit **label-salience and spatial position feature** rather than a noise detector. Appending g_i provides the linear classifier with explicit entity-salience cues, aiding classification when underlying text tokens are severely degraded.

11 Discussion

Our experiments provide three key methodological takeaways for robust PEFT research: 1. **De-coupling Noise Detection from Task Performance**: Supervised noise detection (Auxiliary loss) optimizes corruption identification (AUC 0.91) at the expense of task F1. Conversely, effective task features (GateConcat) ignore synthetic noise status entirely (AUC 0.52). 2. **The Critical Role of Capacity Controls**: Adding scalar pathways increases classifier parameterization. Constant-gate controls ($g \equiv 0.5$) demonstrate that linear head capacity accounts for part of reported robustness gains. Gating claims must be validated against capacity-matched baselines. 3. **Negative Utility of Static Down-Weighting**: Heuristic token masking (SALLoRA) degrades performance because corrupted tokens still contain critical contextual and positional evidence needed for sequence labeling.

11.1 Limitations

This study has several limitations. First, the main experiments use one dataset (FUNSD) and one pretrained backbone (LayoutLMv3). Second, FUNSD is relatively small, and five random seeds provide limited statistical power for separating modest effects. Third, the corruptions are synthetic and may not reproduce the distribution of naturally occurring OCR or scanning errors. Fourth, the GateConcat–Frozen comparison is not statistically conclusive. Finally, the gate analysis is based on a limited number of analyzed runs and should be expanded before making strong claims about the representation learned by the gate.

12 Conclusion

Through a controlled 5-seed study on LayoutLMv3 and FUNSD, we show that hand-crafted quality weighting (SALLoRA) fails to improve document NER under input corruption. While GateConcat provides modest, statistically significant F1 improvements under severe synthetic text noise (+1.51 pp on substitution, +2.30 pp on OCR confusion), capacity controls demonstrate that a constant extra classifier dimension captures part of this gain. Furthermore, token analysis reveals the learned gate acts as a label-salience feature rather than a corruption detector. We conclude that PEFT robustness claims require capacity-matched controls and direct mechanistic analysis to distinguish genuine corruption routing from classifier capacity expansion.

Reproducibility Statement

All experiments use publicly available code and models for LayoutLMv3 and LoRA. The corruption-generation scripts, training configurations, and per-seed results are included in the repository to enable exact reproduction of all reported tables and statistical tests.

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